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24MA313

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Mathematics

III Semester B.E. Degree: TEST-2

ENGINEERING MATHEMATICS-III

Duration: 1 Hour 15 Minutes | Date: 7-01-2026 | Max. Marks:30

NOTE: PART-A is compulsory.
PART-B have internal choice.

Branch: CSE & ISE

PART-A

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO 4	1,2	1.2.1	Find the number of permutations of $a, b, c, \cdots x, y, z$ in which none of the patterns **SPIN**, **GAME**, **PATH** or **NET** occurs.	07
2.	CO 5	1,2	1.2.1	An apple, a banana, a mango and an orange are to be distributed to four boys B1, B2, B3, B4. The boys B1 and B2 do not wish to have apple, the boy B3 does not want banana or mango and B4 refuses orange. In how many ways the distribution can be made so that no boy is displeased?	08

PART-B

Q.NO	CO	CD	PI	QUESTION	MARKS
3.	CO 4	1,2	1.2.1	Find the number of permutations of the letters of the word **INSTITUTION** (i)How many of these are begin with I? *(ii)*How many of these are begin with I and end with *N (iii)* In how many arrangements three *T*'s are together (iv)In how many arrangements three *T*'s are not together	08
4.	CO 5	1,3	1.2.1	In how many ways can the integers 1, 2,3,.....10 be arranged in a line so that no even integer is in its natural place.	07
				OR	
5.	CO 4	1,3	1.2.1	How many distinct four-digit integers can one make from the digits 1, 3, 3, 7, 7 and 8?	08

6	CO 5	1,2	1.2.1	A machine that inserts letters into envelopes goes haywire and insert letters randomly into envelopes. What is the probability that in a group of 100 letters (i)no letter is put into the correct envelope? (ii)exactly 1 letter is put into the correct envelope? (iii)exactly 98 letters are put into the correct envelope? (iv)exactly 99 letters are put into the correct envelope? (v) all letters are put into the correct envelopes?	07

Course Outcome: At the end of the course the students will have the ability to	
CO 4	Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly
CO 5	Derive the generating function for the given situation and evaluate the required coefficient.

PI's	
1.2.1	Apply laws of natural science to an engineering problem

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